

Awesom Living — AI Features: 15-Day Backend Plan

What we need from Atul — deliverables, data shapes, and sample logs

Architecture: Backend gathers data → builds prompt → Claude Sonnet returns text → Backend stores result.
Ground Rules: (1) Backend decides all facts, AI only writes sentences. (2) AI never makes changes to settings/data. (3) One AI on/off toggle per Home. (4) Correlation only — never say one thing *caused* another.

15-Day Timeline Overview

Days	Phase	Deliverables
1 – 2	Prerequisites	AI on/off toggle, alerts.resolution_reason, single alerts migration, verify GLK sleep fields, door/window event list
3 – 4	Feature 1: Daily Narrative	Cron job + Claude prompt, daily_narratives + resident_narratives collections, fallback to plain numbers
5 – 6	Feature 4: Monthly Review & Compare Months	monthly-review + monthly-trend endpoints, classification logic, edge-case handling (no BP, first month)
7 – 8	Feature 2: Ask Abhi on Alerts & Feature 8: Graphs	alert explanation endpoint + caching, graph explanation endpoint + unified analysis shape, context_hash caching
9 – 11	Feature 3: Ask Abhi Chatbot	Fixed tool list (5 tools), conversation storage, ownership validation, date-range cap, clarification flow
12 – 13	Feature 5: Doctor Report	Severity-ranked findings, Claude prose per finding, PDF generation, device disclaimer, doctor_reports collection
14	Feature 6: Threshold Advisor & Feature 7: Insight Feed	Weekly cron for threshold suggestions, dashboard reorder suggestions, trigger rules, prerequisite #4 wired up
15	Integration Testing & Bug Fixes	End-to-end test all 8 features, AI toggle on/off verified, fallbacks confirmed, edge cases covered

Existing MongoDB Collections (already available)

These collections are already in the database. Atul reads from them — does not modify their schema unless noted in prerequisites.

sleep_sessions

```
{ "_id": "ObjectId", "resident_id": "ObjectId",  
  "start_time": "2026-10-14T23:12:00Z", "end_time": "2026-10-15T06:52:00Z",  
  "platform_date": "2026-10-14", "status": "complete", "duration_minutes": 460,  
  "stages": { "light_minutes": 210, "deep_minutes": 120, "rem_minutes": 90, "awake_minutes": 40 },  
  "avg_heart_rate": 64, "avg_respiration": 15, "movement_event_count": 12,  
  "snoring_detected": true, "toss_turn_count": 8 }
```

platform_date = calendar date person went to bed (2 AM on Oct 15 -> platform_date Oct 14).

bp_readings

```
{ "_id": "ObjectId", "home_id": "ObjectId", "resident_id": "ObjectId",  
  "systolic": 142, "diastolic": 88, "pulse": 72,  
  "taken_at": "2026-10-15T08:30:00Z", "received_at": "2026-10-15T08:31:12Z",  
  "mapped_at": "2026-10-15T09:00:00Z", "mapped_by": "ObjectId" }
```

Readings arrive unclaimed. Family member claims (maps) a reading to a Resident via the app.

room_occupancy_state + room_occupancy_events

```
// room_occupancy_state (current snapshot)  
{ "resident_id": "ObjectId", "room_id": "ObjectId", "occupied": true,  
  "since": "2026-10-15T20:12:00Z" }  
// room_occupancy_events (historical log)  
{ "resident_id": "ObjectId", "room_id": "ObjectId",  
  "event": "entered", "timestamp": "2026-10-15T20:12:00Z" }
```

alerts

```
{ "_id": "ObjectId", "resident_id": "ObjectId", "home_id": "ObjectId",  
  "alert_type": "no_motion", "triggered_at": "2026-10-15T20:47:00Z",  
  "resolved_at": "2026-10-15T21:05:00Z", "status": "resolved",  
  "resolution_reason": "confirmed_concern",  
  "context": { "room": "bathroom", "threshold_minutes": 30 } }
```

resolution_reason: false_positive | confirmed_concern — needed by Feature 6.

room_alert_settings

```
{ "resident_id": "ObjectId", "room_id": "ObjectId",  
  "long_stay_threshold_minutes": 45, "no_motion_threshold_minutes": 30 }
```

monthly_summary

```
{ "resident_id": "ObjectId", "month": "2026-10",  
  "avg_sleep_duration_minutes": 440, "avg_heart_rate": 65, "avg_respiration": 15,  
  "avg_bp_systolic": 128, "avg_bp_diastolic": 82, "bp_reading_count": 6,  
  "avg_movement_events": 10, "snoring_nights_count": 4, "avg_toss_turn_count": 7,  
  "alert_counts_by_type": { "no_motion": 1, "long_stay": 0, "fall": 0 }, "total_alerts": 1 }
```

Pre-computed per Resident per calendar month.

dashboard_section_views

```
{ "account_holder_id": "ObjectId", "home_id": "ObjectId",  
  "section": "motion_tracking", "viewed_at": "2026-10-15T21:00:00Z" }
```

Day-by-Day Plan — Features, Data Shapes & Sample Logs

DAYS 1 – 2 : Prerequisites (foundation for all features)

#	What	Blocks	Details
1	AI on/off toggle per Home	ALL 8 features	One toggle in Home settings. When off: data cards show numbers only, "Ask Abhi" buttons hidden throughout the app.
2	alerts.resolution_reason	F2, F6	When closing an alert: must save false_positive or confirmed_concern. Cannot be empty or null.
3	Single alerts table migration	F2, F6	If any alert type still writes to a separate table, migrate everything to unified alerts collection.
4	Verify GLK sleep fields	F5, F8	Confirm GLK actually sends movement_event_count, snoring_detected, toss_turn_count. Confirm data types match schema above.
5	Door/window event list	F3	A log of gate opens/closes separate from occupancy detection. Chatbot tool get_door_window_events reads from this.
6	Customise Dashboard suggested-order	F7	Customise Dashboard must support opening with a proposed (unsaved) starting order passed from a suggestion.

DELIVERABLE: All 6 prerequisites done and testable. AI toggle turns all Abhi features on/off. resolution_reason saved on every alert close. GLK fields confirmed in writing.

DAYS 3 – 4 : Feature 1 — Daily Narrative

What:
Evening card on Dashboard. One headline per Home, one sentence + status pill + stat chips per Resident.

Source:
sleep_sessions (by platform_date), bp_readings (if claimed that day), room_occupancy_state/events, alerts, device status.

New Collections:
daily_narratives, resident_narratives.

API:
Nightly cron job generates. One Claude call per Home (all Residents at once). Headline is templated (not AI).

Fallback:
If AI fails: plain numbers — "Sleep: 7h 40m. Blood pressure: 142/88."

Backend pre-decides:
flags[] per Resident (short_sleep, bp_out_of_range, occupancy_alert, emergency_pressed). resident_status = attention if any flag, else ok.

Sample — daily_narratives document:

```
{ "_id": "ObjectId("6716a...)", "home_id": "ObjectId("670e0...)",  
  "platform_date": "2026-10-15", "generated_at": "2026-10-15T21:30:00Z",  
  "headline": "1 of 2 residents may need a closer look tonight",  
  "status": "generated" }
```

Sample — resident_narratives document:

```
{ "_id": "ObjectId("6716b...)", "daily_narrative_id": "ObjectId("6716a...)",  
  "resident_id": "ObjectId("670f1...)", "resident_status": "attention",  
  "narrative_text": "Blood pressure was a little higher than her average this morning.  
    Otherwise a quiet, normal day.",  
  "flags": ["bp_out_of_range"],  
  "stats": {  
    "sleep": { "value": "6h 10m", "flag": false },  
    "blood_pressure": { "value": "142/88", "flag": true },  
    "activity": { "value": "Normal", "flag": false },  
    "alerts": { "value": 0, "flag": false }  
  } }
```

DELIVERABLE: Cron job runs nightly, stores daily_narratives + resident_narratives. Dashboard card shows headline, per-Resident sentence, status pill, stat chips. Fallback works when AI is down.

DAYS 5 – 6 : Feature 4 — Monthly Review & Compare Months

What:
Two screens per Resident. Monthly Review: headline + 4 stat tiles (sleep, resting_hr, bp, alerts) with better/same/watch. Compare Months: same metrics across last 5 months.

Source:
monthly_summary (already exists, pre-computed from sleep_sessions, bp_readings, alerts).

New Collections:
None — monthly_summary already has everything.

API:
GET /api/residents/{resident_id}/monthly-review?month=2026-10 | GET /api/residents/{resident_id}/monthly-trend?metric=sleep&months=5

Classification rules (Backend decides, NOT AI):
Sleep: better if above last month or own avg, watch if clearly below. Resting HR: same if +/-3 bpm. BP: same if in range. Alerts: better if count down, watch if up.

Edge cases:
No BP readings -> return nothing (not zero). Fewer than 5 months -> return however many exist. First month -> no comparison_label.

Sample — Monthly Review API response:

```
// GET /api/residents/670f1.../monthly-review?month=2026-10
{ "month_label": "October 2026",
  "headline": "A steady month -- sleep and activity stayed close to his usual pattern.",
  "stats": [
    { "key": "sleep", "value": "7h 20m", "comparison_label": "Up from 7h 05m",
      "classification": "better" },
    { "key": "resting_hr", "value": "65 bpm", "comparison_label": "Same as last month",
      "classification": "same" },
    { "key": "bp", "value": "128/82", "comparison_label": "Stable, 6 readings",
      "classification": "same" },
    { "key": "alerts", "value": "3", "comparison_label": "Down from 5",
      "classification": "better" }
  ] }
```

Sample — Compare Months API response:

```
// GET /api/residents/670f1.../monthly-trend?metric=sleep&months=5
{ "metric": "sleep", "months": [
  { "label": "Jun", "value": "6h 48m" }, { "label": "Jul", "value": "7h 00m" },
  { "label": "Aug", "value": "7h 10m" }, { "label": "Sep", "value": "7h 05m" },
  { "label": "Oct", "value": "7h 20m", "is_current": true }
], "note": "Trending up over the last two months." }
```

DELIVERABLE: Both endpoints working with correct classification logic. Edge cases handled (no BP, first month, <5 months). One Claude call generates the headline text.

DAYS 7 – 8 : Feature 2 — Ask Abhi on Alerts + Feature 8 — Ask Abhi on Graphs

Feature 2 — Ask Abhi on Alerts

What:

Button on every alert card. Explains this one alert + shows 30-day history pattern for same alert_type + Resident.

New Collection:

alert_explanations (cached per alert — made once, reused on every view).

API:

POST /api/alerts/{alert_id}/explanation

Backend pre-computes:

total_count, most_common_time_of_day, most_common_room, trend (going_up|stable|going_down), resolved vs open counts. AI gets these aggregated numbers, never raw alert list.

Fallback:

Show plain stats with retry button for AI text.

Sample — API response:

```
// POST /api/alerts/6714c.../explanation
{ "status": "ready",
  "explanation": "No movement detected for 35 min after entering bathroom at 8:12 PM.",
  "historical_analysis": "4th no-motion alert in last 30 days, mostly 8-9 PM.",
  "overall_note": "Nothing unusual vs the evening pattern -- worth a check-in.",
  "stats": { "count_last_30_days": 4, "most_common_time": "8 to 9 PM", "trend": "stable" } }
```

Sample — alert_explanations document:

```
{ "_id": "ObjectId('6717a...)", "alert_id": "ObjectId('6714c...)",
  "explanation": "No movement detected for 35 min after...",
  "historical_analysis": "4th no-motion alert in last 30 days...",
  "overall_note": "Nothing unusual vs the evening pattern...",
  "generated_at": "2026-10-15T21:10:00Z" }
```

Feature 8 — Ask Abhi on Graphs

What:

Button on every graph in the app. Explains that graph in plain words and flags anything noteworthy.

New Collection:

graph_explanations (cached by context_hash — same night reuses saved answer).

API:

POST /api/graphs/explanation

Unified analysis shape:

One backend function per graph_type returns: summary_stats, baseline_comparison, is_significant, flags. AI called identically for all graph types.

Sample — API request + response:

```
// POST /api/graphs/explanation
// Request:
{ "graph_type": "sleep_stage_timeline", "resident_id": "670f1...",
  "context": { "date": "2026-10-14" } }
// Response:
{ "simple_explanation": "Mostly light sleep, shorter deep sleep block around 2 AM.",
  "flag_text": "Deep sleep ~65 min, lower than usual 90 min.",
  "flags": [ { "type": "low_deep_sleep" } ] }
```

Sample — graph_explanations document:

```
{ "_id": "ObjectId('671ba...)", "graph_type": "sleep_stage_timeline",
  "resident_id": "ObjectId('670f1...)", "context_hash": "a3f8c2...d41e",
  "simple_explanation": "Mostly light sleep, shorter deep sleep...",
  "flag_text": "Deep sleep ~65 min, lower than usual 90 min.",
  "generated_at": "2026-10-15T07:00:00Z" }
```

DELIVERABLE: Both Features 2 and 8 working. Explanations cached — second view loads instantly. Fallback shows plain stats. Graph types: sleep_stage_timeline, bp_trend, monthly_metric_trend.

DAYS 9 – 11 : Feature 3 — Ask Abhi Chatbot

What:

FAB button on Dashboard -> chat screen. Free-text questions about Residents/Homes.

New Collection:

conversations (chat history per Home).

API:

POST /api/homes/{home_id}/chat (send message) | GET /api/homes/{home_id}/chat/recent (load last conversation)

Fixed Tool List (AI picks tool + params, Backend executes safe query):

Tool	Reads From	Parameters
get_sleep_summary	sleep_sessions	resident_id, start_date, end_date
get_bp_readings	bp_readings	resident_id, start_date, end_date
get_occupancy_summary	room_occupancy_state/events	resident_id, room (opt), start_date, end_date
get_door_window_events	door/window event list	home_id, device_name (opt), start_date, end_date
get_alerts	alerts	resident_id or home_id, alert_type (opt), start_date, end_date

Security:

AI never writes DB queries directly. Backend validates resident_id/home_id ownership. Date ranges capped at 12 months.

Sample — Chat request + response:

```
// POST /api/homes/670e0.../chat
// Request:
{ "message": "How has sleep been for Rishabh this month?", "conversation_id": "conv_abc123" }
// Response (answered):
{ "conversation_id": "conv_abc123", "status": "answered",
  "reply": "Rishabh averaged 7h 15m of sleep this month, close to his usual range.
           3 nights below 6 hours, all within the past two weeks.",
  "source_summary": "Based on 30 nights of sleep data" }
// Response (needs_clarification):
{ "status": "needs_clarification", "reply": "Did you mean Rishabh or Meera?",
  "clarification_options": ["Rishabh", "Meera"] }
```

DELIVERABLE: Chat endpoint working with all 5 tools. Conversation history saved and loadable. Ownership validated. Date ranges capped. Clarification flow handles ambiguous questions.

DAYS 12 – 13 : Feature 5 — Doctor Report

What:

Side Menu -> "Generate report with Abhi." One-month PDF for a Resident, meant for their doctor.

Source:

monthly_summary (reuse Feature 4 data) + sleep_sessions (movement/snoring/toss fields).

New Collection:

doctor_reports (stores metadata + PDF link).

API:

POST /api/residents/{resident_id}/doctor-report

Finding severity:

bp_out_of_range = 3, sleep_decline = 2, unresolved_alerts = 3, sleep_quality_pattern = 2. Backend ranks, AI writes prose per finding.

Critical rule:

Correlation language ONLY — "happened together" or "may be related," NEVER "caused." Backend appends device disclaimer AFTER AI finishes.

Sample — doctor_reports document:

```
{ "_id": "ObjectId('6718a...)", "resident_id": "ObjectId('670f1...)",
  "period_start": "2026-10-01", "period_end": "2026-10-31",
  "generated_at": "2026-11-01T10:15:00Z", "status": "ready",
  "findings": [
    { "type": "bp_out_of_range", "severity": 3,
      "detail": "4 of 6 readings above 140/90 this month." },
    { "type": "sleep_quality_pattern", "severity": 2,
      "detail": "Deep sleep averaged 25 min below 30-day baseline." }
  ],
  "pdf_url": "https://storage.awesomeliving.com/reports/6718a...pdf" }
```

DELIVERABLE: Doctor report PDF generated on demand. Findings ranked by severity. Correlation language enforced. Device disclaimer appended. PDF stored and URL saved in doctor_reports collection.

DAY 14 : Feature 6 — Threshold Advisor + Feature 7 — Insight Feed

Feature 6 — Adaptive Threshold Advisor

What:

Weekly suggestion card. Suggests loosening or tightening no_motion/long_stay threshold based on alert history.

New Collection:

threshold_suggestions.

Trigger rules:

Loosen: 5+ false_positive alerts, none confirmed_concern. Tighten: 2+ confirmed_concern where time-before-confirm < current setting.

Runs:

Once a week. Never triggered by a single alert closing. Prerequisite: alerts.resolution_reason must be populated.

Sample — threshold_suggestions document:

```
{ "_id": "ObjectId('6719a...)", "resident_id": "ObjectId('670f1...)",
```

```
"alert_type": "no_motion", "direction": "loosen",
"current_threshold_minutes": 30, "suggested_threshold_minutes": 40,
"supporting_incident_ids": ["ObjectId("6714c...)", "ObjectId("6714d...)",
"ObjectId("6714e...)", "ObjectId("6714f...)", "ObjectId("67150...)",
"status": "pending", "generated_at": "2026-10-20T03:00:00Z" }
```

Feature 7 — Personalised Insight Feed

What:
Suggests reordering Dashboard sections based on which detail pages the Account Holder opens most.

New Collection:
dashboard_suggestions.

Trigger rule:
top_section by view count NOT already first in saved order AND count >= 2x second section.

Sample — dashboard_suggestions document:

```
{ "_id": "ObjectId("671aa...)", "account_holder_id": "ObjectId("670d8...)",
"home_id": "ObjectId("670e0...)",
"proposed_order": ["motion_tracking", "sleep", "vitals", "alerts", "camera"],
"supporting_counts": { "motion_tracking": 12, "sleep": 5, "vitals": 4 },
"status": "pending", "generated_at": "2026-10-21T03:00:00Z" }
```

DELIVERABLE: Both weekly crons running. Threshold suggestions generated with correct trigger rules. Dashboard suggestions generated. Status flow: pending -> dismissed | applied.

DAY 15 : Integration Testing & Bug Fixes

- Test every feature end-to-end. Specific checks:**
- AI toggle OFF -> all Abhi UI hidden, no Claude calls made, plain data still shows.
 - AI toggle ON -> all 8 features produce output with real data.
 - Fallbacks: kill Claude endpoint -> Daily Narrative shows plain numbers, Alert explanation shows stats + retry.
 - Chatbot security: try accessing another Home's Resident -> must return 403.
 - Doctor Report: verify "caused" never appears in AI output. Device disclaimer present at bottom.
 - Monthly Review edge cases: Resident with 0 BP readings, Resident with only 1 month of data.
 - Threshold Advisor: verify it does NOT fire on a single alert close.
 - Graph explanations: same night same graph -> cached response returned (no new Claude call).

DELIVERABLE: All 8 features working end-to-end. All fallbacks verified. All edge cases covered. Ready for frontend integration.

New API Endpoints — Complete List

Endpoint	Method	Feature	Day
/api/alerts/{alert_id}/explanation	POST	2: Ask Abhi on Alerts	7-8
/api/homes/{home_id}/chat	POST	3: Chatbot	9-11
/api/homes/{home_id}/chat/recent	GET	3: Chatbot	9-11
/api/residents/{id}/monthly-review	GET	4: Monthly Review	5-6
/api/residents/{id}/monthly-trend	GET	4: Compare Months	5-6
/api/residents/{id}/doctor-report	POST	5: Doctor Report	12-13
/api/graphs/explanation	POST	8: Graphs	7-8

New Collections — Complete List

Collection	Feature	Purpose	Day
daily_narratives	1: Daily Narrative	One per Home per evening	3-4
resident_narratives	1: Daily Narrative	Per-Resident under daily_narratives	3-4
alert_explanations	2: Alerts	Cached explanation per alert	7-8
conversations	3: Chatbot	Chat history per Home	9-11
doctor_reports	5: Doctor Report	Report metadata + PDF links	12-13
threshold_suggestions	6: Threshold	Pending/dismissed/applied	14
dashboard_suggestions	7: Insight Feed	Dashboard reorder suggestions	14
graph_explanations	8: Graphs	Cached graph explanations	7-8